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Social Media Privacy Guard: Context-Aware AI Anonymization

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Problem Statement

Background

- Social media users unknowingly expose sensitive personal information — faces, license plates, identity documents, house numbers, and personal text — in publicly shared images and videos.

Core Problem

- Traditional privacy tools apply blanket blurring to all detected objects, degrading image quality and removing important visual context that users want to preserve.
- Existing systems lack context-awareness: they blur regardless of whether the face is the main subject, a background person, or even an image on a poster.

Research Gap

- No current system combines context-aware selective anonymization with Privacy Risk Scoring, user-controlled rules, and local privacy-first processing in a single unified pipeline.

Proposed Solution

- The Social Media Privacy Guard system uses deep learning (YOLOv8, CNN, OCR) to detect sensitive objects and intelligently anonymizes only relevant background elements, preserving the primary subject of the image while generating a measurable Privacy Risk Score (0-100).

Literature Survey

Sl no	Author and Published Year	Paper Title	Methodology Used	Problem Solved	Advantages	Inference and Limitations
1.	A. Frome et al. (2009) IEEE ICCV	Large-Scale Privacy Protection in Google Street View	Sliding-window face detector with OCR-based license plate detection; two-stage high-recall pipeline	Automated blurring of faces and license plates in Google Street View	High recall (89% faces, 94-96% plates); domain-specific post-processing	Domain-specific; daytime clear conditions only; no deep learning used
2.	J. Redmon et al. (2016) IEEE CVPR	You Only Look Once: Unified, Real-Time Object Detection (YOLO)	Single CNN predicting bounding boxes and class probabilities simultaneously over grid	Real-time unified object detection across 80+ classes at high speed	1000x faster than R-CNN; global context awareness; single-pass pipeline	Struggles with small/grouped objects; lower precision than two-stage detectors
3.	K. Zhang et al. (2016) IEEE Signal Processing Letters	Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks (MTCNN)	Three-stage cascaded CNN with online hard sample mining for joint detection and alignment	Accurate face detection and landmark localization in unconstrained environments	Real-time; superior accuracy on FDDB and WIDER FACE benchmarks	Performance degrades with extreme occlusions or very low-resolution faces
4.	H. Hukkelås & F. Lindseth (2023) IEEE WACV	DeepPrivacy2: Towards Realistic Full-Body Anonymization	Style-based GAN with CSE-guided synthesis for full-body and face anonymization	Realistic anonymization of full human figures beyond face-only blurring	Stronger privacy guarantees; high image quality; editable diverse outputs	Computationally expensive; requires pose estimation and dense surface embeddings

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5.	Y. Baek et al. (2019) IEEE CVPR	Character Region Awareness for Text Detection (CRAFT)	CNN generating character region score and affinity maps to detect text in arbitrary shapes	Robust text detection in natural scenes with curved, tilted, or artistic fonts	Excels on arbitrary-shaped text; state-of-the-art on ICDAR benchmarks	Slower than simpler detectors; may miss extremely small or low-contrast text
6.	H. Li et al. (2018) IEEE Trans. ITS	Toward End-to-End Car License Plate Detection and Recognition With Deep Neural Networks	Unified deep neural network jointly localizing plates and recognizing characters in a single forward pass	Eliminates intermediate errors in separate detection-then-recognition pipeline	End-to-end trainable; faster processing; multi-dataset evaluation	Performance drops on distorted or occluded plates; limited multilingual support
7.	S. Ren et al. (2016) IEEE TPAMI	Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks	Region Proposal Network sharing convolutional features with object detection head	Near real-time object detection by eliminating the selective search bottleneck	State-of-the-art accuracy; unified end-to-end training; efficient shared features	Slower than single-shot detectors (SSD, YOLO); two-stage pipeline complexity
8.	P. Viola & M. Jones (2001) IEEE CVPR	Rapid Object Detection Using a Boosted Cascade of Simple Features (Haar Cascade)	AdaBoost classifier cascade over Haar-like features on integral images	First real-time face detection system enabling practical everyday deployment	Extremely fast; computationally light; pioneered real-time face detection	High false-positive rate; poor performance with occlusion and non-frontal faces

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9.	A. Dosovitskiy et al. (2021) ICLR	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT)	Pure transformer applied to fixed-size image patches for image classification	Demonstrates vision transformers can replace CNNs with sufficient training data	Excellent scalability; outperforms CNNs on large datasets; strong representation	Requires very large datasets; computationally expensive; less inductive bias than CNNs
10.	A.G. Howard et al. (2017) Google/arXiv	MobileNets: Efficient CNNs for Mobile Vision Applications	Depthwise separable convolutions drastically reducing computation and model parameters	Enabling real-time deep learning inference on mobile and embedded/edge devices	8-9x computation reduction; suitable for on-device privacy-first local processing	Accuracy-speed tradeoff; lower performance on complex multi-class detection tasks
11.	D. Karatzas et al. (2015) IEEE ICDAR	ICDAR 2015 Competition on Robust Reading (Scene Text Benchmark)	Standardized evaluation dataset and metrics for text detection and recognition in scene images	Provides unified benchmark for comparing OCR and text detection methods	Widely adopted standard; enables fair comparison across OCR approaches	English-only; controlled conditions; not representative of social media image diversity
12.	R. Smith (2007) IEEE ICDAR	An Overview of the Tesseract OCR Engine	Adaptive thresholding, connected components analysis with LSTM-based recognition	Open-source OCR for extracting printed and typed text from images	Highly accurate on printed text; supports 100+ languages; free and open source	Struggles with stylized, curved, or degraded scene text in social media photos

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13.	D. Qi et al. (2021) ArXiv/IEEE	YOLO5Face: Why Reinventing a Face Detector	YOLOv5 fine-tuned specifically for multi-scale face detection with landmark regression	High-performance real-time face detection for privacy and surveillance systems	Lightweight yet accurate; multi-scale FPN; supports tiny face detection	Requires domain-specific fine-tuning; limited facial landmark precision vs MTCNN
14.	B. Shi et al. (2017) IEEE TPAMI	An End-to-End Trainable Neural Network for Image-Based Sequence Recognition (CRNN)	CNN for feature extraction + Bi-LSTM for sequence modeling + CTC for transcription	Scene text recognition without requiring character-level segmentation	End-to-end trainable; handles variable-length text; robust to fonts and styles	Requires clean text region crops; limited on heavily distorted or curved text
15.	X. Zhou et al. (2017) IEEE CVPR	EAST: An Efficient and Accurate Scene Text Detector	Fully convolutional network predicting text geometry directly from feature maps	Fast and accurate text detection in natural scenes with arbitrary orientations	Extremely fast inference (13.2 fps); minimal post-processing pipeline	Less effective on very long text lines; requires NMS post-processing
16.	M. Maximov et al. (2020) IEEE CVPR	CIAGAN: Conditional Identity Anonymization Generative Adversarial Networks	Conditional GAN replacing face identity while preserving appearance attributes	Anonymizing images/videos while maintaining utility for downstream detection tasks	Realistic anonymized output; downstream task performance preserved	Limited to face region; GAN training instability; identity leakage risk

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17.	D. Ibdah et al. (2021) IEEE Access	Why Should I Read the Privacy Policy, I Just Need the Service	User survey and behavioral analysis of digital privacy attitudes on online platforms	Reveals gap between user privacy needs and actual policy comprehension	Highlights urgent need for automated context-aware privacy tools	Self-reported survey data; limited to policy comprehension, not image privacy
18.	G. Jocher et al. (2023) Ultralytics	Ultralytics YOLOv8: Real-Time Object Detection, Segmentation and Classification	Anchor-free multi-task architecture: detection, segmentation, pose estimation in one model	Unified real-time framework for object detection and face detection in privacy systems	Best accuracy-speed tradeoff; Python API; multi-task in a single model	Newer architecture with limited formal peer-reviewed benchmarks at release
19.	H. Yang et al. (2024) IEEE Trans. IFS	G2Face: High-Fidelity Reversible Face Anonymization via Generative and Geometric Priors	GAN combining generative priors and 3D geometric priors for reversible face anonymization	High-quality face replacement that is both privacy-preserving and visually realistic	Reversible for authorized access; superior clarity over blur/pixelation	Complex training; not suitable for real-time edge deployment
20.	H. Kung et al. (2025) IEEE WACV	Face Anonymization Made Simple	Diffusion-based face replacement preserving expression, gaze and pose while altering identity	Simplified but powerful face anonymization resistant to re-identification attacks	Preserves emotional expressiveness; resists re-identification; minimal training needed	Slower inference per image than blur; requires diffusion model compute resources

Research Gaps Identified

Gap 1: Blanket Blurring Without Context Awareness

- Existing systems (Google Street View, dashcam anonymizers) blur every detected face/plate uniformly, destroying important image context and aesthetics of social media content.

Gap 2: No Main Subject Preservation

- Current tools fail to distinguish between the primary subject (user's own face) and background individuals, resulting in unacceptable photo quality for social media sharing.

Gap 3: Reversible Anonymization Vulnerability

- Gaussian blur and pixelation can be reversed by modern AI (diffusion models achieve 95.9% re-identification accuracy on blurred faces), making standard blurring methods inadequate.

Gap 4: Limited OCR for Artistic/Scene Text

- Standard OCR tools miss stylized, tilted, or artistically rendered text common on T-shirts, posters, and banners in social media photos.

Gap 5: No Privacy Risk Quantification

- No existing tool provides a measurable, explainable privacy risk score to inform users of their actual exposure level before sharing content.

Finalized Objectives

Objective 1: Intelligent Sensitive Object Detection

- Detect faces, license plates, identity documents, house numbers, and sensitive text using YOLOv8 and CNN-based detection with high precision and recall.

Objective 2: Context-Aware Selective Anonymization

- Implement main-subject detection to preserve the primary portrait subject while selectively anonymizing only background sensitive elements.

Objective 3: Privacy Risk Scoring

- Develop a Privacy Risk Score (0-100) based on number, type, and visibility of detected sensitive objects to quantify exposure levels.

Objective 4: Robust OCR-Based Text Privacy

- Integrate CRAFT + EasyOCR pipeline to detect and anonymize PII text (emails, phone numbers, addresses) in stylized, curved, and scene text.

Objective 5: User-Controlled Privacy Rules

- Provide a user interface allowing selective toggling of anonymization categories (faces, plates, documents, text) with adjustable blur strength.

Objective 6: Privacy-First Local Processing

- Ensure all processing occurs on-device with no cloud upload, preserving user data sovereignty.

Proposed Methodology

Stage 1: Input Processing

- Accept user-uploaded image/video; perform preprocessing (resize, normalize) for model inference.

Stage 2: Multi-Model Detection Engine

- YOLOv8 (fine-tuned): Detects faces, license plates, and personal documents with bounding boxes.
- CRAFT + EasyOCR: Detects and reads scene text including stylized and curved text for PII identification.
- MTCNN / DeepFace: Secondary face detection for improved recall in complex scenes.

Stage 3: Context Analysis

- Main Subject Check: If face bounding box covers >30% of frame, classify as main subject and exempt from blurring.
- Confidence Filtering: Ignore low-confidence detections to minimize false positives.

Stage 4: Anonymization Layer

- Apply user-selected anonymization: Gaussian blur, pixelation, emoji overlay, or generative face inpainting (GAN/Diffusion-based).

Stage 5: Privacy Risk Scoring

- Compute Privacy Risk Score (0-100) based on count, type, and visibility of sensitive regions.

Stage 6: Output

- Return anonymized image with Explainable Highlight Mode (labeled bounding boxes) and before/after comparison view.

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